

Energy Context & Modeling Scenarios: Thailand as a Case Study

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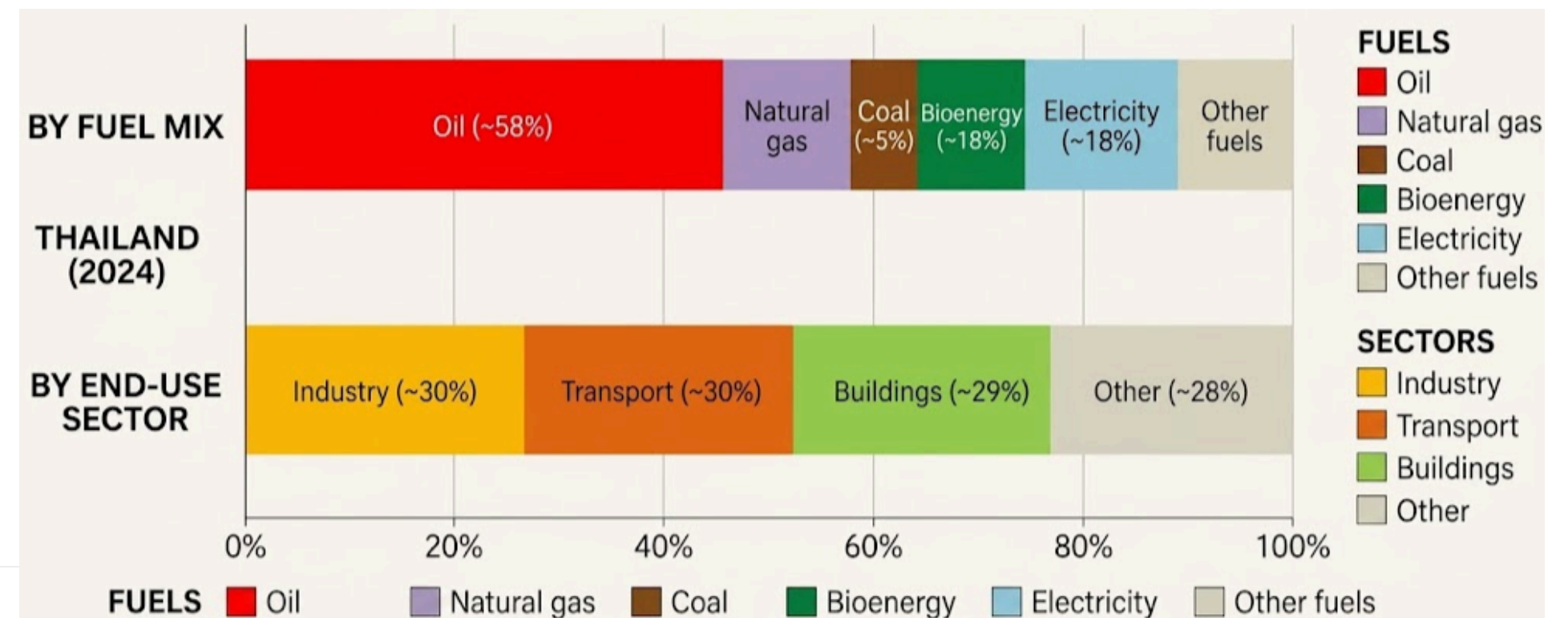
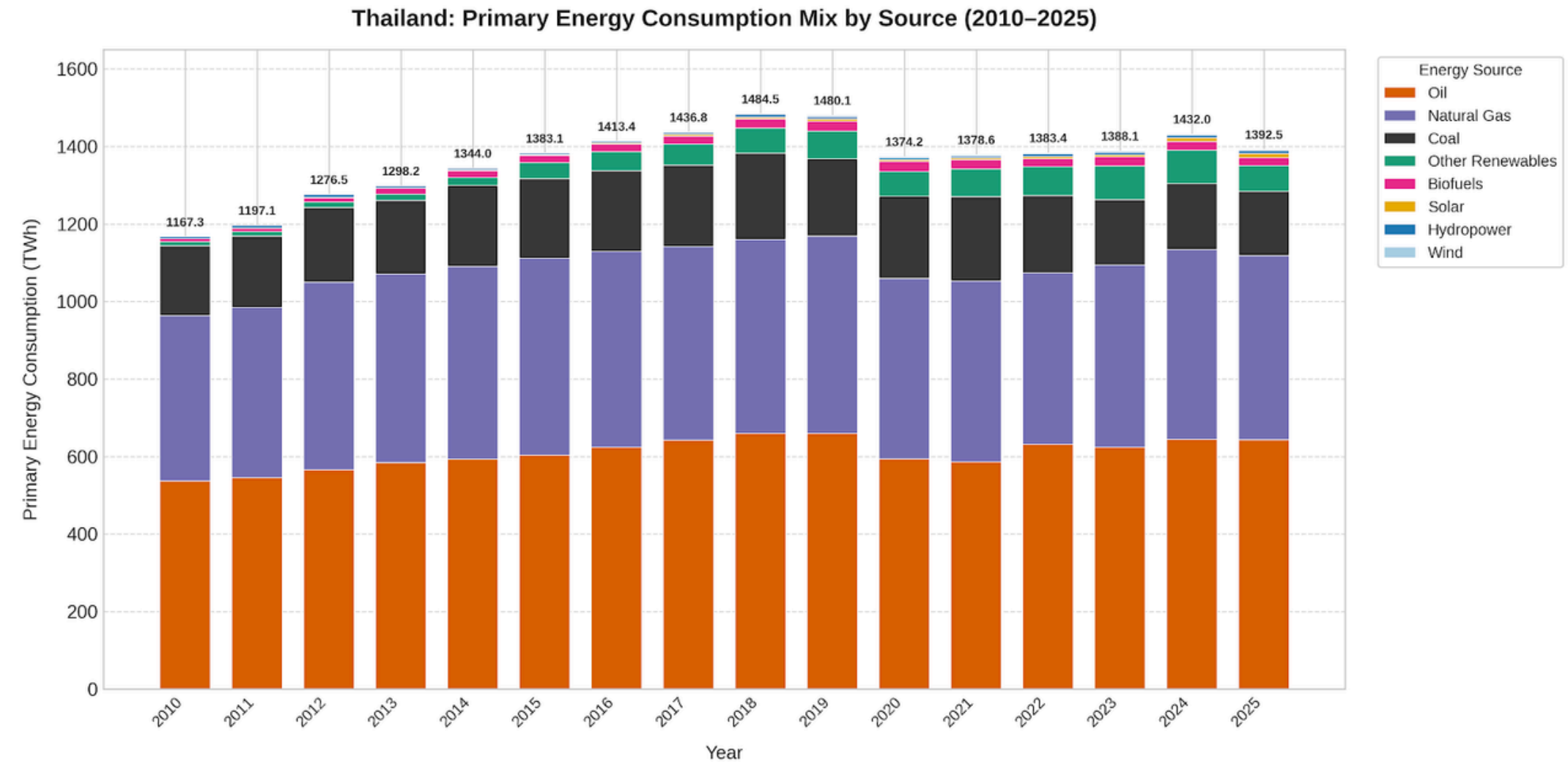


Energy Context

Nationally Determined Contribution (NDC) commits to achieve net-zero GHG by 2025 and reduce net emissions by 47% below 2019 levels for 2031-2035.

Battery Energy Storage Systems (BESS) & Smart Grids are emerging into traditional grid system to support NDC and secure the grid stability.

Sources: Our World in data;
International Energy Agency. (2026). Southeast Asia Energy Outlook 2026



Proposed Modeling Scenarios

Scenario 1

Demand Response Integration with Peak Load Shifting

Objective: Implement automated and incentive-based Demand Response programs targeting residential sector to flatten peak electricity demand.

Scenario 2

Integrated Demand Response with Distributed Solar & BESS

Objective: Combine Demand Response capability with utility-scale and rooftop solar PV paired with Battery Energy Storage Systems (BESS).